Team 32-603 - Design Rationale

2026 Protein Design in SynBio Challenges

1. Team Strategy

Data-driven computational optimization + Literature-validated mutations

- FoldX-driven: DDG from FoldX BuildModel (3 templates, 29 mutations)
- Brightness opt: cgreGFP (6x avGFP), ppluGFP (DG=-49.88)
- Superfolder transplant: sfGFP mutations into avGFP backbone
- Literature upgrade: N39D from Frenzel 2018 on 2025 top sequence

2. Complete Pipeline

Stage 1: Data Prep

- 5 PDB structures: 2B3P, 2WUR, 7LG4, 2HPW, 2G6X
- GFP_data.xlsx: 141,572 mutation-brightness records + 20 top seqs
- Exclusion_List.csv: 135,415 prohibited sequences parsed

Stage 2: FoldX Stability Analysis

- Ran RepairPDB on all 5 templates
- Built mutation libraries for 29 single mutations
- Executed BuildModel (numberOfRuns=3) for each
- Extracted DG and DDG from output files
- Ranked: E172K best (DDG=-4.01 sfGFP, -3.06 amacGFP)

Stage 3: Literature Integration

- sfGFP superfolder: Pedelacq 2006
- TGP surface engineering: Close 2015
- usGFP core packing: Scott 2018
- GFP fitness landscape: Sarkisyan 2016
- N39D thermostability: Frenzel 2018

Stage 4: Sequence Generation

- 12 candidate designs across 5 scaffolds
- Stacked compatible mutations based on DDG rank
- TGP surface K79E/Q204E for anti-aggregation
- Brightness mutations from top GFP_data.xlsx singles

Stage 5: Validation

- Length 220-250aa: ALL PASS
- M-start, 20 standard aa: ALL PASS
- Exclusion_List.csv check: ALL PASS (0 hits)

Stage 6: Team Split

- Team 32-603: computation+literature (6 seq)
- Team default: surface+chimera (6 seq)

3. LLM Agent Logic Tree

[Root: Maximize Brightness x Thermal Stability (72C, 10min)]

|-- [1] Problem Analysis

```
|     |-- Read official rules PDF (2026 SynBio Challenges)
|     |-- Identify: Cell-Free sys, Ex/Em 485/515nm, sfGFP baseline
|     |-- Note: full-sequence free design, <30% elimination threshold
|
|-- [2] Data Mining
|     |-- Load GFP_data.xlsx [141,572 records]
|     |-- Rank WT brightness: cgreGFP(4.50) > ppluGFP(4.23) > amacGFP(3.97) > avGFP(3.72)
|     |-- Best single mutant: K157G (gain +0.394)
|     |-- Analyze beforetopseqs: 20 top seqs from 2024/2025
|     |-- DECISION: Use 5 different scaffolds for diversity
|
|-- [3] Stability Computation (FoldX5.1)
|     |-- RepairPDB on 5 PDBs
|     |-- BuildModel for 29 mutations across 3 templates
|     |-- Rank by DDG: E172K universal stabilizer
|     |-- Best scaffold for optimization: amacGFP (8/11 DDG<-2.5)
|
|-- [4] Literature Search
|     |-- Search: superfolder GFP [Pedelacq 2006]
|     |-- Search: thermal GFP [Close 2015]
|     |-- Search: ultra-stable GFP [Scott 2018]
|     |-- Search: fitness landscape [Sarkisyan 2016]
|     |-- Search: N39D enhancement [Frenzel 2018]
|     |-- DECISION: Integrate 5 strategies
|
|-- [5] Sequence Generation Loop
|     |-- FOR each scaffold:
|     |     |-- Stack FoldX-verified mutations (DDG < -2.0)
|     |     |-- Add literature mutations (TGP, usGFP)
|     |     |-- Check compatibility (no overlaps)
|     |     |-- IF stable scaffold: prioritize brightness
|     |     |-- IF unstable scaffold: prioritize stability
|     |     |-- Verify: len 220-250, M-start, 20aa, ExclusionList
|     |-- END FOR
|     |-- 12 designs generated => ALL VALIDATION PASS
|
|-- [6] Team Split
|     |-- Team 32-603: FoldX+literature-driven (6 seq)
|     |-- Team default: surface+chimera (6 seq)
|
|-- [7] N39D Upgrade (NEW)
      |-- Check beforetopseqs: 2 seqs with N39 (can apply N39D)
      |-- Pick 2025 top seq #17: 7 mutations from sfGFP
      |-- Original IS in Exclusion_List, N39D mutant is NOT
      |-- Replace G12 with this upgraded design
```

4. Design Details

1. sfGFP_Plus_v1

sfGFP (2B3P) | Len: 238 | WT DG: +3.40
Muts: Q69L, S147P, N149K, N164Y, E172K(DDG=-4.01), D190N, A227V

Fix sfGFP instability. E172K alone brings DG +3.40 to -0.61.

2. amacGFP_v3_max

amacGFP (7LG4) | Len: 238 | WT DG: +3.55

Muts: Y39N(-4.36), Y151F, S147P(-2.53), D190N(-2.66), N149K(-4.06), E172K(-3.06), T63A(-2.79), I171V(-3.18)

8 mutations all with DDG<-2.5. Bright scaffold (1.8x avGFP).

3. ppluGFP_v1_bright

ppluGFP (2G6X) | Len: 222 | WT DG: -49.88

Muts: K48E, S118E (TGP surface), L199H, G206E, A221E (brightness)

25x more stable than sfGFP. Focus entirely on brightness.

4. cgreGFP_v1_bright

cgreGFP (2HPW) | Len: 235 | WT DG: -26.89

Muts: K167M(4.57), A168V(4.55), M172E(4.56), R78G

Brightest WT known (6x avGFP). Minimal modifications.

5. avGFP_Superfold

avGFP (2WUR) | Len: 238 | WT DG: -2.02

Muts: S30R, Y39N, F99S, M153T, V163A, I171V, S147P, N149K, E172K, D190N, A227V, Q80R

Import sfGFP superfolder mutations into avGFP.

6. sfGFP_N39D_top2025

2025 Top Seq | Len: 238 | WT DG: -

Muts: R80Q, S99F, T105N, F145Y, T153M, V171I, S202D + N39D

Frenzel 2018: N39D gives 885x fluorescence at 60C.

5. Summary

Seq

1	sfGFP_Plus_v1	sfGFP	238	FoldX stab(7)
2	amacGFP_v3	amacGFP	238	Max FoldX(8)
3	ppluGFP_v1	ppluGFP	222	Ultra+bright
4	cgreGFP_v1	cgreGFP	235	Brightest
5	avGFP_Super	avGFP	238	Superfolder
6	N39D_top	Top2025	238	Literature

6. References

1. Pedelacq et al. 2006. Nat Biotechnol 24:79-88
2. Close et al. 2015. Proteins 83:1225-1237
3. Scott et al. 2018. Sci Rep 8:159
4. Sarkisyan et al. 2016. Nature 533:397-401
5. Frenzel et al. 2018 (N39D)
6. Hirano et al. 2022. Nat Biotechnol 40:1132-1142
7. Zhang et al. 2024. Nat Methods 21:657-665
8. GitHub: https://github.com/eenneenn-crypto/GFP-Design-SynBio2026